

Homework 6
COMS 3310
Due Tuesday April 28th, 11:59pm

No late work accepted.

Please read the HW guidelines (see next page). Failure to follow the guidelines will result in a lower grade.

1. Design a Turing Machine for each of the following languages using JFLAP, using the indicated filenames, and submit them all to **Homework 6 TMs** on Gradescope. The autograder for this part of the Gradescope assignment will be available latest by Thursday evening.

Please click on <https://www.jflap.org/tutorial/> and click on **Turing Machine** and then **Construct and Run** for guidance on how to design a TM. When asked what type of TM to create, choose **Standard Turing Machine**.

(a) **hw6q1.jff**:

$$L_1 = \{a^n b^n c^n \mid n \geq 0\}$$

(b) **hw6q2.jff**:

Let $|x|_a$ denote the number of occurrences of the symbol a in the string x . Design a TM for the following language over $\Sigma = \{a, b\}$.

$$L_2 = \{x \mid |x|_a = |x|_b\}$$

(c) **hw6q3.jff**:

Let L_3 be a language over the alphabet $\{a, b\}$, defined inductively as follows (incidentally, this is a language of balanced parentheses, in which a and b are open and close parentheses of the same type).

- $\epsilon \in L$.
- If $x \in L$, then axb is in L .
- If x and y are in L , then xy is in L .

2. Practice Problems – Do Not Submit

You may discuss these problems with each other, or with a teaching assistant or instructor at office hours.

- (a) Prove that the class of Turing-acceptable (that is, Turing-recognizable) languages is closed under intersection. For this problem do we need to run a TM for a limited number of steps?
- (b) Prove that the class of Turing-acceptable languages is closed under concatenation. For this problem do we need to run a TM for a limited number of steps?

Guidelines:

- Please submit your HW via Gradescope to the autograder.
- You are allowed to collaborate with *others*, while doing your homework.
- Definition of *others*: Students who are taking the course this semester.
- You can consult GenAI tools.
- However, you should write the final solutions alone, without consulting anyone. Your writing should demonstrate that you understand the proofs completely.
- For each problem, you should acknowledge all the individuals you consulted and whether you used any GenAI resources. This will not affect your grade. Failure to acknowledge is considered *academic dishonesty*, and it will affect your grade.
- When proofs are required, you should make them both clear and rigorous. Do not hand wave.
- Any regrading requests should be made within one week of returning the homework. No requests will be entertained after this one week period.